

- (a)** Explain why it is important for plants to open and close their stomata in a daily rhythm.

..... [3]

-
- open stomata / %
- time / hours
- Key
- day
- night
- | Time (hours) | Open Stomata (%) | Period |
|--------------|------------------|--------|
| 0 | 15 | Day |
| 6 | 90 | Day |
| 12 | 75 | Day |
| 18 | 45 | Day |
| 24 | 25 | Night |
| 30 | 75 | Day |
| 36 | 90 | Day |
| 42 | 55 | Day |
| 48 | 35 | Night |
| 54 | 80 | Day |
| 60 | 70 | Day |
| 66 | 50 | Day |

Fig. 2.1

Fig. 2.1 shows that the percentage of open stomata increases in the first seven hours of the experiment. Describe the sequence of changes that occurs in the guard cells that leads to the stomata opening.

[illegible]

(c) The experiment was repeated with *A. thaliana* plants that were left in darkness from 14 to 96 hours. The results are shown in Fig. 2.2.



..... [2]

[Total: 11]